MIL-T-48895
7 July 1977
SUPERSEDING
See Section 6

MILITARY SPECIFICATION

TEST SET, ATOMIC WEAPON: T4197

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of test set entitled Test Set, Atomic Weapon: T4197 which is used with the Sprint Warhead.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-202 - Test Methods for Electronic and Electrical Equipment.
MIL-STD-1168- Lot Numbering of Ammunition.
MIL-STD-1460- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure For.

DRAWINGS

ORDNANCE CORPS

9258169 - Circuit Card Assembly.
9272368 - Panel, Blank.
9272377 - Panel Assembly.
9272716 - Box, Fiberboard, Packaging, For Test Set T-4197.
9273595 - Test Set, Atomic Weapon: T-4197.
EL-9244084 - Equipment List.
EL-9258169 - Equipment List.
EL-9269233 - Equipment List.
EL-9269234 - Equipment List.
EL-9269235 - Equipment List.

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document, should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAN-Q, Dover, New Jersey 07801, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.

FSC 1190

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:43Z
Check the source to verify that this is the current version before use.

EL-9272368 - Equipment List.
EL-9272377 - Equipment List.
EL-9273595 - Equipment List.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The test set shall comply with all requirements specified on the drawing (dwg) 9273595, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. The contractor shall assign serial numbers as required by drawings. The contractor may select any number which complies with the drawing requirement. These numbers shall not be duplicated.

3.5 Performance. The test set, drawing 9273595, shall operate as described in 4.4.1 without physical damage (see 6.5).

3.6 Electrical characteristics.

3.6.1 Circuit resistance. The permissible circuit resistance of panel shall be as indicated in Table I prior to conformal coating. The resistance shall be measured as described in 4.4.2.1.

Table I

In accordance with drawing 9272377

From	To	Resistance (In Ohms)
J1-A	S1B-10	0.1 max
J1-B	S1A-10	0.1 max
J1-C	S1A-8	0.1 max
J1-D	S1B-8	0.1 max
J1-E	S1A-7	0.1 max
J1-F	S1B-7	0.1 max

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:43Z
Check the source to verify that this is the current version before use.

From	To	Resistance (In Ohms)
J2-5	TB-E14	0.1 max
J2-7	TB-E16	0.1 max
J2-66	TB-E16	0.1 max
J2-67	TB-E13	0.1 max
J2-6	TB-E15	0.1 max
J2-8	TB-E15	0.1 max
J3-Y	TB-E12	0.1 max
J3-D	TB-E11	0.1 max
J3-B	TB-E11	0.1 max
J3-A	TB-E12	0.1 max
G1-1	TB-E5	0.1 max
G1-2	TB-E7	0.1 max
J4-4	TB-E13	0.1 max
J4-5	TB-E14	0.1 max
DS1-1	TB-E6	0.1 max
DS1-2	TB-E2	0.1 max
DS1-3	TB-E4	0.1 max
S1A-1	All points (Switch in Position 2)	Greater than 9 megohms
S1A-2	All points	Greater than 9 megohms
S1A-3	S1B-3	0.1 max
S1A-4	TB-E8	0.1 max
S1A-5	TB-E18	0.1 max
S1A-9	S1A-10	0.1 max

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:43Z
Check the source to verify that this is the current version before use.

From	To	Resistance (In Ohms)
S1A-11	S1A-8	0.1 max
S1A-12	S1A-7	0.1 max
S1A-13	TB-E1	0.1 max
S1B-1	All points (Switch in Position 2)	Greater than 9 megohms
S1B-2	All points	Greater than 9 megohms
S1B-4	TB-E9	0.1 max
S1B-5	TB-E17	0.1 max
S1B-9	S1B-10	0.1 max
S1B-11	S1B-8	0.1 max
S1B-12	S1B-7	0.1 max
S1B-14	TB-E3	0.1 max
S1A-6	All points	Greater than 9 megohms
S1B-6	All points	Greater than 9 megohms

3.6.2 Continuity and circuit resistance of cable assemblies T4197-1, -2 and -3 and lead, electrical. There shall be electrical continuity between points listed in Table II, with a circuit resistance of not more than 0.20 ohm maximum for cables T4197-1, -2 and -3, and 0.05 ohm maximum for lead, electrical. Testing shall be as specified in 4.4.2.2.

Table II

Cable T4197-1 in accordance with drawing 9269233

From	To
P1-C	P3-A
P1-D	P3-B

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:43Z
Check the source to verify that this is the current version before use.

From	To
P1-E	P3-Y
P1-F	P3-D
P1 Shell (Shield)	P3 Shell (Shield)

Cable T4197-2 in accordance with drawing 9269234

From	To
P2-5	P1-A
P2-6	P1-B
P2-7	P1-C
P2-8	P1-D
P2-66	P1-E
P2-67	P1-F
P2 Shell (Shield)	P1 Shell (Shield)

Cable T4197-3 in accordance with drawing 9269235

From	To
P1-A	P4-4
A1-B	P4-5
P1 Shell (Shield)	P4 Shell (Shield)

Lead, electrical in accordance with drawing 9244084.

From	To
Terminal lug	Connector, Plug, Electrical

3.6.3 Circuit resistance of circuit card assembly prior to installing into panel assembly. The circuit resistance between points listed in Table III shall be as specified when tested in accordance with 4.4.2.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:43Z
Check the source to verify that this is the current version before use.

Table III

In accordance with drawing 9258169

From	To	Resistance (Ohms)
E17	E18	4K ± 1%
E11	E12	2K ± 1%
E3	E4	4K ± 1%
E1	E2	3.4K ± 1%
E2	Junction of Resistor #3 and CR2	1K ± 1%

3.6.4 Insulation resistance of cable assemblies T4197-1, -2 and -3. The permissible resistance shall be 50 megohms (meg) minimum (min) between points listed in Table IV. The resistance shall be measured as described in 4.4.2.4.

Table IV

Assembly Drawing 9269233 - Cable T4197-1

Connector - From	To - (Tied Together)
P3, each pin, for all pins	P3, all other pins, and shell.

Assembly Drawing 9269234 - Cable T4197-2

Connector - From	To - (Tied Together)
P2, each pin, for all pins	P2, all other pins, and shell.

Assembly Drawing 9269235 - Cable T4197-3

Connector - From	To - (Tied Together)
P4, each pin, for all pins	P4, all other pins, and shell.

3.7 Physical characteristics. The test set shall withstand impact shock and transportation vibration without physical damage (see 6.5). Testing shall be as described in 4.4.3. After vibration, the test set shall be tested and conform to 3.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:43Z
Check the source to verify that this is the current version before use.

3.7.1 Impact shock. The test set shall be subjected to 15 ± 0.5 gravity units (g), peak (sine) three times in each direction along each of its three mutually perpendicular axes (18 shocks total). Each acceleration shall last 11 ± 1 millisecond (ms) in duration.

3.7.2 Transportation vibration. The test set shall be subjected to the parameters specified in the vibration schedule, Table V along each of its three mutually perpendicular axes.

Table V

Vibration Schedule
Amplitude

Frequency Hertz (Hz)	Gravity Units (peak)	Double Amplitude (inches)	Sweep Time ± 1/2 minute
5-10-5		0.08 ± 0.008	7.5
11-15-11	0.4 ± 0.04		7.5
16-50-16		0.036 ± 0.0036	7.5
51-500-51	5.0 ± 0.5		7.5
		Total Time Along Each Axis	30 ± 2 min

3.8 Environmental characteristics. The test set shall withstand high, low temperature, and humidity without physical damage (see 6.5). Testing shall be as described in 4.4.4. After humidity, the test set shall be tested and conform to 3.5; then examined for damage.

3.8.1 High temperature. The test set shall be exposed to a temperature of 160 ± 5 degrees Fahrenheit (°F) for a minimum of four hours and until the temperature of the test set has stabilized. Tests shall be in accordance with 4.4.4.1.

3.8.2 Low temperature. The test set shall be exposed to a temperature of -65 ± 5°F for four hours minimum and until the temperature of the test set has stabilized. Tests shall be in accordance with 4.4.4.2.

3.8.3 Humidity. The test set shall be exposed to four 24 hour temperature humidity cycles at temperature ranging between 150 ± 5°F and 70 ± 5°F with relative humidity (RH) maintained at 95 +5, -0 percent. Tests shall be in accordance with 4.4.4.3.

3.9 Soldering. Soldering shall be in accordance with MIL-STD-1460.

3.10 Workmanship.

3.10.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:43Z
Check the source to verify that this is the current version before use.

3.10.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.10.3 Burrs. Parts shall be free from burrs.

3.10.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.10.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.10.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 1 complete assembly (see dwg 9273595) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table VI. Tests shall be conducted in order specified in Table VI.

Table VI
First article inspections and tests

Inspection or Test	Dest	Non-Dest
1. Classification of defects/ characteristics (see 4.3.2.1)		X
2. Circuit resistance (see 3.6.1)		X

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:43Z
Check the source to verify that this is the current version before use.

Inspection or Test	Dest	Non-Dest
3. Continuity and circuit resistance of cable assemblies (see 3.6.2)		X
4. Circuit resistance of circuit card assembly (see 3.6.3)		X
5. Insulation resistance of cable assemblies (see 3.6.4)		X
6. Performance (see 3.5)		X
7. Impact shock (see 3.7.1)	X	
8. Transportation vibration (see 3.7.2)	X	
9. High temperature (see 3.8.1)		X
10. Low temperature (see 3.8.2)		X
11. Humidity (see 3.8.3)	X	

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor defects/characteristics table (see Table VII and Table VIII), and inspected for all defects/characteristics in the applicable classification. Any defects/characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of defects/characteristics in each classification need not be followed as long as all defects/characteristics inspections are performed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:43Z
Check the source to verify that this is the current version before use.

Table VII
Major Defects/Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

Table VIII
Minor Defects/Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

4.3.2.1 Classification of defects/characteristics (see 3.2). Critical, major and minor defects/characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Cable assemblies T4197-1, -2 & -3 prior to soldering shield (see dwgs 9269233, 9269234 and 9269235 covering details of dwg 9273595).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Connector pins/sockets missing, cracked or flattened	Visual
102.	Connector inserts cracked or punctured	Visual
103.	Shields/braid wire incomplete or missing	Visual
104.	Insulation sleeving/shrinkable tubing damaged or missing	Visual
105.	Adapters/reducers cracked or missing	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.10)	Visual

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:43Z
Check the source to verify that this is the current version before use.

4.3.2.1.2 Cable assemblies T4197-1, -2 & -3 prior to potting (see dwgs 9269233, 9269234 and 9269235).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Soldering improper (see MIL-STD-1460)	Visual
102.	Wires improperly crimped to connector	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.10)	Visual

4.3.2.1.3 Cable assemblies T4197-1, -2 & -3 (see dwgs 9269233, 9269234 and 9269235).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Potting improper	Visual
102.	Failure of connectors to engage mating member	Manual/Gage
103.	Components damaged or improperly assembled	Visual
104.	Sleeving damaged exposing braid/wiring	Visual
105.	Lengths incorrect	Gage
Minor:		
201.	Markings missing, incorrect, illegible or misplaced	Visual
202.	Lacquer over markings missing or insufficient	Visual
203.	Evidence of poor workmanship (see 3.10)	Visual
204.	Objective evidence that cable assemblies have passed electrical tests	Visual

4.3.2.1.4 Circuit card assembly prior to assembling into panel assembly (see dwg 9258169 covering a detail of dwg 9272377).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:43Z
Check the source to verify that this is the current version before use.

Major:

101.	Diodes and thyristor not of correct value and type	Visual
102.	Diodes and thyristor improperly oriented	Visual
103.	Nylon washer missing from underneath thyristor	Visual
104.	Diodes/thyristor damaged or cracked	Visual
105.	Soldering improper (see MIL-STD-1460)	Visual
106.	Delamination of land or terminal areas from board	Visual
107.	Conductors peeling or not bonded to board	Visual
*108.	Possible shorts between conductors (bridging)	Visual/Gage
*109.	Splits or cracks across the conductors	Visual/Gage
110.	Land and terminal areas not completely covered by plating	Visual

Minor:

201.	Markings missing, incorrect or illegible	Visual
202.	Evidence of poor workmanship (see 3.10)	Visual

*NOTE: This inspection shall be performed using an optical instrument of 7 X minimum magnification.

4.3.2.1.5 Panel prior to installing components (see dwg 9272368).

Categories	Defects/Characteristics	Method of Inspection

Critical: None defined.

Major:

101.	Mislocation of mounting holes for generator (4 holes)	Gage
102.	Mislocation of small locking holes for switch shaft (1 hole)	Gage
103.	Connector holes improperly oriented (4 holes)	Visual

Minor:

201.	Markings missing, improper or illegible	Visual
202.	Lacquer missing or insufficient on panel face	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:43Z
Check the source to verify that this is the current version before use.

4.3.2.1.6 Case (see dwg 9244081 covering a detail of dwg 9273595).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Objective quality evidence that case conforms to specification and drawing is missing, improper, or incomplete	Visual
	102. Components missing, improper, damaged, misplaced, loose, improperly assembled, installed (inside and outside)	Visual
	103. Latch (2) do not unlatch and latch freely	Manual/Visual
Minor:		
	201. Evidence of poor workmanship (see 3.10)	Visual
	202. Finish improper	Visual

4.3.2.1.7 Lead, electrical (see dwg 9244084 covering a detail of dwg 9273595).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Pin not peened over (see note on dwg)	Visual
	102. Wire braids damaged	Visual
	103. Cable overall length incorrect	Gage
	104. Cable connector plug or terminal lug improper	Visual
	105. Soldering improper (see MIL-STD-1460)	Visual
	106. Terminal lug improperly crimped to wire braid	Visual/Manual
	107. Objective evidence that cable has passed electrical test	Visual
Minor:	None defined.	

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:43Z
Check the source to verify that this is the current version before use.

4.3.2.1.8 Test Set, Atomic Weapon: T4197 (see dwg 9273595).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Cable assemblies damaged or missing	Visual
102.	Connector receptacles damaged	Visual
103.	Receptacle connector caps missing	Visual
104.	Rotary switch, generator knob or light indicator inoperable	Manual
105.	Any part damaged, loose or improperly assembled	Visual/Manual
106.	Finish damaged exposing base material	Visual
107.	R.F. gasket missing	Visual
108.	Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
201.	Identification plate/markings missing, improper or illegible	Visual
202.	Lacquer over markings missing or insufficient	Visual
203.	Evidence of poor workmanship (see 3.10)	Visual

4.3.2.1.9 Preparation for delivery before test set (see dwg 9273595) is packaged.

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Components missing, damaged, loose or improper (case open)	Manual/Visual
102.	Protective cover on connectors loose or missing (case open)	Manual/Visual
103.	Lid over cables loose (case open)	Manual/Visual
104.	Components missing, damaged or loose (case closed)	Manual/Visual
Minor:		
201.	Marking obliterated	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:43Z
Check the source to verify that this is the current version before use.

4.3.2.1.10 Preparation for delivery (see 5).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Packaging, packing, or marking does not conform to dwg 9272716 and the contract	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.10)	Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test Classification
a. Electrical characteristics	Major
b. Performance	Major

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Electrical characteristics, and performance. Each test set shall be tested in accordance with 4.4.2 and 4.4.1. If any item fails to comply with 3.6 and 3.5, the item shall be rejected and removed from the lot.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major defects/characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see 2.1). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:43Z
Check the source to verify that this is the current version before use.

4.3.4.2 Minor defects/characteristics and tests. Acceptance inspection equipment for minor defects/characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Performance testing. The test set, firmly positioned and with appropriate cables connected shall be operated in the following sequence and in the manner described.

A. Pre-Test Unit Verification.
B. Pre-Test Cable Verification - Sprint T4197-1 Cable.
C. Post-Test Cable Verification - Sprint T4197-1 Cable.
D. Pre-Test Cable Verification - Spartan T4197-2 Cable.
E. Post-Test Cable Verification - Spartan T4197-2 Cable.
F. Pre-Test Cable Verification - Sprint T4197-3 Cable.
G. Post-Test Cable Verification - Sprint T4197-3 Cable.
H. Post-Test Unit Verification.

A. Pre-Test Unit Verification:

1. Open Cover of Test Set.
2. Turn S1 on Test Set to 1-OFF.
3. Depress DS1, while rotating generator knob. DS1 should light.
4. Turn S1 to 3 - GEN AND LAMP TEST.
5. Turn generator knob. DS1 should light.
6. Turn S1 to 4 - SELF TEST.
7. Turn generator knob. DS1 should light.
8. Turn S1 to 5 - DISC TEST.
9. Turn generator knob. DS1 should not light.
10. Depress DS1, while rotating generator knob. DS1 should light.
11. Turn S1 on Test Set to 1-OFF.

B. Pre-Test Cable Verification - Sprint T4197-1 Cable:

1. Open lid of test set cover containing T4197-1 Sprint Cable.
2. Remove T4197-1 cable and close lid.
3. Remove dust covers from J1-COMMON and J3-SPRINT on panel of test set.
4. Remove dust covers from P1 and P3 ends of T4197-1 Cable.
5. Attach P1 end of cable to J1 on panel.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:43Z
Check the source to verify that this is the current version before use.

6. Attach P3 end of cable to J3 on panel.
7. Turn S1 to 7 - SPRINT B.
8. Turn generator knob. DS1 should light.
9. Turn S1 to 8 - SPRINT C.
10. Turn generator knob. DS1 should light.
11. Turn S1 to 1-OFF.
12. Detach P3 end of cable from panel.

C. Post-Test Cable Verification - Sprint T4197-1 Cable:

1. Attach P3 end of T4197-1 Sprint Cable to J3 on panel.
2. Turn S1 to 7 - SPRINT B.
3. Turn generator knob. DS1 should light.
4. Turn S1 to 8 - SPRINT C.
5. Turn generator knob. DS1 should light.
6. Turn S1 to 1-OFF.
7. Detach P3 end of Cable from J3 on panel.
8. Detach P1 end of Cable from J1.
9. Place dust covers on P1 and P3 ends of cable.
10. Place dust covers on J1 and J3 on panel.
11. Open lid of test set cover which is to contain T4197-1 Sprint cable.
12. Place cable inside and close lid.

D. Pre-Test Cable Verification - Spartan T4197-2 Cable:

1. Open lid of test set cover containing T4197-2 Spartan cable.
2. Remove cable and close lid.
3. Remove dust covers from J1-COMMON and J2-SPARTAN on panel of test set.
4. Remove dust covers from P1 and P2 ends of cable.
5. Attach P1 end of cable to J1 on panel.
6. Attach P2 end of cable to J2 on panel.
7. Turn S1 to 10 - SPARTAN A.
8. Turn generator knob. DS1 should light.
9. Turn S1 to 11 - SPARTAN B.
10. Turn generator knob. DS1 should light.
11. Turn S1 to 12 - SPARTAN C.
12. Turn generator knob. DS1 should light.
13. Turn S1 to 1-OFF.
14. Detach P2 end of cable from J2 on panel.

E. Post-Test Cable Verification - Spartan T4197-2 Cable:

1. Attach P2 end of T4197-2 Spartan cable to J2 on panel.
2. Turn S1 to 10 - SPARTAN A.
3. Turn generator knob. DS1 should light.
4. Turn S1 to 11 - SPARTAN B.
5. Turn generator knob. DS1 should light.
6. Turn S1 to 12 - SPARTAN C.

Source: https://assist.dla.mil -- Downloaded: 202640-22T19:43Z
Check the source to verify that this is the current version before use.

7. Turn generator knob. DS1 should light.
8. Turn S1 to 1-OFF.
9. Detach P2 end of cable from J2 on panel.
10. Detach P1 end of cable from J1 on panel.
11. Place dust covers on P1 and P2 ends of cable.
12. Place dust covers on J1 and J2 on panel.
13. Open lid of test set cover which is to contain T4197-2 Spartan cable.
14. Place cable inside and close lid.

F. Pre-Test Cable Verification - SPRINT T4197-3 Cable:

1. Open lid of test set cover containing T4197-3 Sprint Cable.
2. Remove cable and close lid.
3. Remove dust covers from J1-COMMON and J4-SPRINT on panel of test set.
4. Remove dust covers from P1 and P4 ends of cable.
5. Attach P1 end of cable to J1 on panel.
6. Attach P4 end of cable to J4 on panel.
7. Turn S1 to 9 - SPRINT D.
8. Turn generator knob. DS1 should light.
9. Turn S1 to 1-OFF.
10. Detach P4 end of cable from panel.

G. Post-Test Cable Verification - Sprint T4197-3 Cable:

1. Attach P4 end of T4197-3 Sprint cable to J4 on panel.
2. Turn S1 to 9 - SPRINT D.
3. Turn generator knob. DS1 should light.
4. Turn S1 to 1-OFF.
5. Detach P4 end of cable from J4 on panel.
6. Detach P1 end of cable from J1 on panel.
7. Place dust covers on P1 and P4 ends of cable.
8. Place dust covers on J1 and J4 on panel.
9. Open lid of test set cover which is to contain T4197-3 Sprint cable.
10. Place cable inside and close lid.

H. Post-Test Unit Verification.

1. Turn S1 to 1-OFF.
2. Depress DS1 while rotating generator knob. DS1 should light.
3. Turn S1 to 3 - GEN AND LAMP TEST.
4. Turn generator knob. DS1 should light.
5. Turn S1 to 4 - SELF TEST.
6. Turn generator knob. DS1 should light.
7. Turn S1 to 5 - DISC TEST.
8. Turn generator knob. DS1 should not light.
9. Depress DS1 while rotating generator knob. DS1 should light.
10. Turn S1 on test set to 1-OFF.
11. Close cover of test set.

Source: https://assist.dla.mil -- Downloaded: 202640-22T19:43Z
Check the source to verify that this is the current version before use.

4.4.2 Electrical testing.

4.4.2.1 Circuit resistance of panel. The resistance between circuits connected in combinations listed in Table I shall be measured with an approved tester (voltmeter-ammeter method). The permissible resistance is indicated under appropriate column in Table I. The measurements shall be made on the complete assembly before conformal coating and before mounting in the case. Tests shall be made with Switch S1 in position 1 OFF, except as noted in Table I.

4.4.2.2 Continuity and circuit resistance of cable assemblies. The resistance of each specified circuit shall be measured with an approved tester in accordance with MIL-STD-202, Method 307 to determine compliance with 3.6.2.

4.4.2.3 Circuit resistance of circuit card assembly. The resistance of each specified circuit shall be measured with an approved tester in accordance with MIL-STD-202, Method 303 to determine compliance with 3.6.3.

4.4.2.4 Insulation resistance of cable assemblies. The insulation resistance between circuits connected in combinations listed in Table IV shall be measured with an approved tester (megohm-meter method). To determine conformance to 3.6.4, a test potential of 500 ± 50 volts direct current (vdc) shall be impressed across each circuit for one minute before a reading is taken. The megohm-meter shall be current limited to 5 ma max.

4.4.3 Physical testing.

4.4.3.1 Impact shock. The test set shall be mounted on an approved fixture. The fixture shall be capable of being oriented to shock the test set in its three axes. The mounted test set then shall be secured to an approved shock device and tested to the requirements in 3.7.1.

4.4.3.2 Transportation vibration. The test set shall be mounted on an approved fixture capable of being oriented to vibrate the test set in its three axes (fig. 1). The mounted test set then shall be secured onto an approved vibration device and tested to the requirements in 3.7.2. After vibration, the test set shall be tested and conform to 3.5.

4.4.4 Environmental testing. The total volume occupied by the test set shall not exceed one-third of the total temperature-chamber volume for conditioning to high low temperature, and humidity.

4.4.4.1 High temperature. The test set shall be placed into an approved temperature chamber and the temperature raised to 160 ± 5°F for four hours minimum and until the temperature of the test set has stabilized (see 6.4).

4.4.4.2 Low temperature. The test set shall be placed in an approved temperature chamber and the temperature lowered to -65 ± 5°F for four hours minimum and until the temperature of the test set has stabilized (see 6.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:43Z
Check the source to verify that this is the current version before use.

4.4.4.3 Humidity. The test set shall be placed in an approved temperature-humidity chamber and subjected to the requirements specified in 3.8.3. Each 24-hour cycle shall be regulated as follows:

(A). Gradually raise internal chamber temperature to 150 + 5°F and the RH not less than 95 percent over a period of two hours.

(B). Maintain condition of step (A) for not less than six hours (total consecutive time: eight hours).

(C). Maintain 95 percent or greater RH and reduce internal chamber temperature uniformly in 16 hours to 70 ± 5°F (total consecutive time: 24 hours, or 1 cycle).

After the test set has been exposed to four humidity cycles (96 consecutive hours), it shall be removed from the temperature chamber and after removal, but within one hour thereafter, the test set shall be tested and conform to 3.5 and examined for damage (see 6.5).

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The test set shall be packed in accordance with dwg 9272716.

5.1.2 Level C. The test set shall be packaged in accordance with dwg 9272716 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The test set shall be packed in accordance with dwg 9272716.

5.3 Marking. Marking shall be as specified in accordance with dwg 9272716.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:43Z
Check the source to verify that this is the current version before use.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAN-I, Dover, New Jersey 07801.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest or lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Physical damage. A defect, detectably by visual examination. A 5X magnifying glass may be used in examining for defects.

(1) External. Illegible marking, damaged finish, exposing base metal to include all hardware and other case shell, evidence of corrosion, dents, crack, split - to include all hardware and outer case shell bent connector pins, broken lamp, inoperable switch.

(2) Internal. Evidence of moisture, component loose, damage, evidence of corrosion.

6.6 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-2316 dated 5 June 1972, SCN No. 1 dated 30 August 1972, SCN No. 2 dated 17 November 1972 and SCN No. 3 dated 30 July 1973.

Custodian:
Army - AR

Preparing activity:
Army - AR

Project No. 1190-A107

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:43Z
Check the source to verify that this is the current version before use.

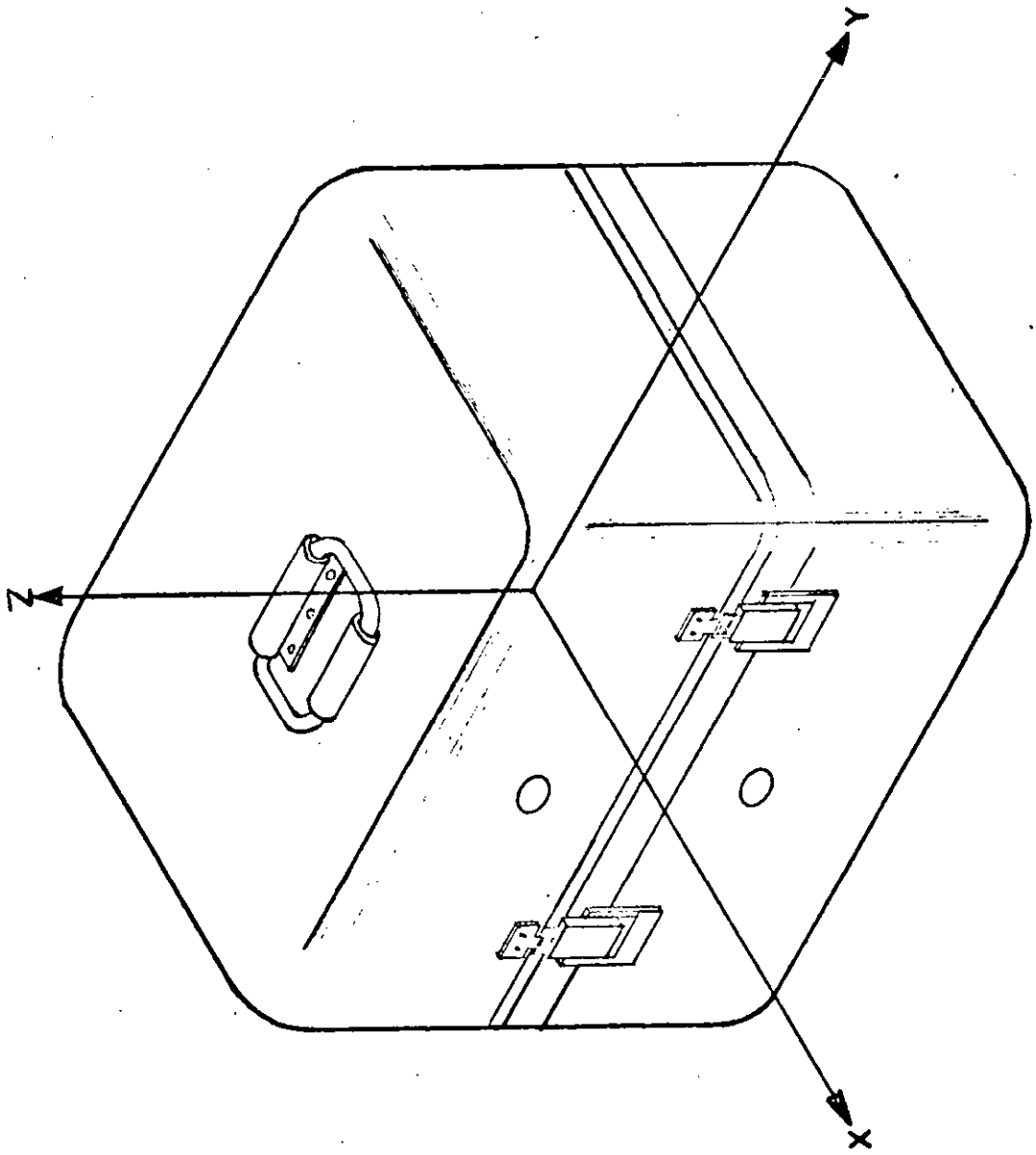


Figure 1. Test Set T4197
(Sealed As If Prepared For Shipment)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:43Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314

U.S.MAIL

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
US Army Armament R&D Command
ATTN: DRDAR-QAN-Q
Dover, NJ 07801

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:43Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:43Z
Check the source to verify that this is the current version before use.